

AYA BOUKARI

Computer Engineering Student | AI & Software Engineering | Agentic AI & RAG

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LinkedIn | GitHub | Portfolio

Profile

Third-year Computer Engineering student at the National Engineering School of Tunis (ENIT), concurrently pursuing a Master's degree in Information Processing and Living Systems Complexity (TICV). Hands-on experience in **AI, Agentic AI, RAG, LLMs, Machine Learning, Edge Computing, and Software Engineering**. Experienced in developing intelligent platforms, microservices architectures, and AI applications for healthcare and hydrometric data management. Seeking a **Final-Year Internship (PFE)** in AI Engineering, Agentic AI, Machine Learning, or Software Engineering.

Education

National Engineering School of Tunis (ENIT) 2024 – Present
Engineering Degree in Computer Engineering – 3rd Year | Ranked **405/1700** in the national entrance examination.

Master's Program in Information Processing and Living Systems Complexity (TICV) – ENIT 2026 – Present
Machine Learning, Signal Processing, Computer Vision, Distributed Systems, Cloud Computing

Preparatory Institute for Engineering Studies of Monastir (IPEIM) 2022 – 2024 | **Baccalaureate in Mathematics** 2022

Experience & Key Projects

Intelligent Hydrometric Data Management Platform *DGRE-ENIT / Summer 2026*

- Developed the **Python backend**, PostgreSQL database, and part of the frontend.
- Designed **Agentic AI workflows** and a **RAG pipeline** for a chatbot answering domain-specific hydrometric questions.
- Automated the generation of **hydrometric yearbooks and maps**.

Technologies: Python, PostgreSQL, LLMs, RAG, Agentic AI, REST APIs, Geospatial Data

IoMT Telemonitoring Platform for Cardiac Remote Monitoring *PFA2 – ENIT RISC Lab / 2025–2026*

- Designed an **event-driven microservices architecture** with Docker, MQTT, and Kafka for PPG/ACC signal processing.
- Implemented an **XGBoost** cardiac anomaly detection model: 14 features, inference time below 50 ms.
- Developed an **Edge Computing** preprocessing layer, secure API Gateway with Keycloak/OAuth2/JWT, and React dashboard.
- Validated the platform with **46,258 messages/hour**, 100% success rate, 340 ms average latency, and 50 simultaneous patients.

Technologies: Java, Spring Boot, FastAPI, Python, Docker, Kafka, MQTT, XGBoost, InfluxDB, PostgreSQL, Keycloak, React

Automatic Absence Seizure Detection from EEG Signals *Data Mining – ENIT / 2025–2026*

- Built an EEG pipeline covering preprocessing, annotation extraction, segmentation, and time/frequency feature engineering using FFT.
- Compared **Decision Tree, Random Forest, KNN, and SVM** on imbalanced data, prioritizing Recall and F1-score.

Technologies: Python, Pandas, NumPy, Scikit-learn, SciPy, Jupyter

Other Projects

Intelligent Nutrition Chatbot – PFA1: LLM fine-tuning + RAG, Flask/React application.

Smart Pen for Dyslexic Users – IEEE: Flutter application, Bluetooth connectivity, speech and learning modules.

International Records Management: Java/Spring Boot backend, REST APIs, Scrum methodology.

Technical Skills

Programming: Python, Java, JavaScript, SQL, HTML, CSS | **AI/ML:** Machine Learning, LLMs, RAG, Agentic AI, XGBoost, Scikit-learn, NLP, Computer Vision, Signal Processing

Backend: FastAPI, Flask, Spring Boot, REST APIs | **Frontend:** React, Flutter | **Databases:** PostgreSQL, InfluxDB, MariaDB

Distributed Systems: Microservices, Kafka, MQTT, Event-Driven Architecture | **DevOps:** Docker, Docker Compose, Git, GitHub, Keycloak, OAuth2, JWT

Activities, Certifications & Languages

IEEE ENIT – Women in Engineering: Treasurer & General Secretary (2024–2025), event and conference organization, **3rd place in the WIE Challenge**.

G2FOSS ENIT: Active Member (2025–2026), open-source hackathons and technical events.

Certifications: CCNA 1, CCNA 2 – Cisco | Efficient LLM Customization

Languages: Arabic (Native) | French (B2) | English (B2) | German (Beginner) | Japanese (Beginner)